

CM 8202
Seminar in New Communication Issues:
Research-based Digital Media for Innovation



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L3

Related Talks

- New Digital Media
- Interactive Digital Multimedia Applications
- Research-based Digital Media for Innovation
- Multidisciplinary between Information Technology and Digital Media

Augmented Reality Magic

Augmented Reality Magic

- Now that an ever-accelerating cascade of eye-popping visual technology such as AR has threatened to steal some of the magic dust from old-fashioned magicians
- Along comes a pasteboard prestidigitator who folds augmented reality into the magic world



Augmented Reality Magic (card trick)

- Blend of technology and illusion creates the magic of tomorrow, today



Integration leads to Innovation

Information
Technology Research
(Eye Tracking)



Comm. Arts
Research
(Advertising
Strategy)



Research
Innovation

Graduate School of Communication Arts and Management Innovation

Oxford Internet Institute

Understanding life online



DPhil in Information, Communication and the Social Sciences

The DPhil in Information, Communication and the Social Sciences (doctoral) programme provides an opportunity for highly qualified students to undertake cutting edge Internet-related research. We are looking for academically excellent candidates who display the potential and enthusiasm necessary to perform research that will make a difference -- to ask important questions and to adopt innovative methodologies and approaches for exploring those questions. Many OII doctoral students are pursuing research that will shape the development of digital networked spaces and those whose lives are affected by it.



OII Research Fellow Kathryn Eccles has been appointed 'Digital Humanities Champion' for the University of Oxford. She will help develop the cross-University Digital Humanities strategy.

<http://www.oii.ox.ac.uk/>



KEIO MEDIA DESIGN.
慶應義塾大学大学院メディアデザイン研究科
GRADUATE SCHOOL OF MEDIA DESIGN, KEIO UNIVERSITY

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Keio University

General Info

Digital media is changing from conventional mass media to personal media that uses media infrastructure for community building. Examples include the convergence of broadcasting and communications and the emergence of social networking services. Content adds value to media, and design enhances all aspects of life. These have become important engines in enriching the human heart and mind. For media, content, and design to take root in society, it is crucial to understand how they function as industries, to understand their business and management models, and to include policies about intellectual property & describe in "Our philosophy of education, " we train "media innovators," defined as people who



Graduate School of Media Design



KEIO MEDIA DESIGN
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<http://www.kmd.keio.ac.jp/>

Power of Pokemon GO

- Computer Vision-based
- Location-based



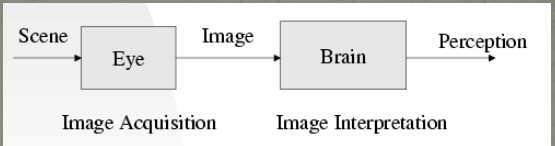



www.forbes.com

Introduction to Computer Vision

Vision

- Vision is the process of discovering what is present in the world and where it is by looking.



```

graph LR
    Scene --> Eye
    Eye -- Image --> Brain
    Brain -- Perception --> Perception
    subgraph Image_Acquisition [Image Acquisition]
        Eye
    end
    subgraph Image_Interpretation [Image Interpretation]
        Brain
    end
  
```

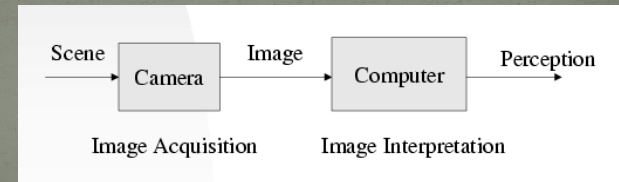

Computer Vision

- “**Computer vision** is the science and technology of machines that see. As a scientific discipline, computer vision is concerned with the theory behind artificial systems that extract information from images. The image data can take many forms, such as video sequences, views from multiple cameras, or multi-dimensional data from a medical scanner.”

By Dana H. Ballard, Computer Vision.

Computer Vision

- Computer Vision is the study of analysis of pictures and videos in order to achieve results similar to those as by men.



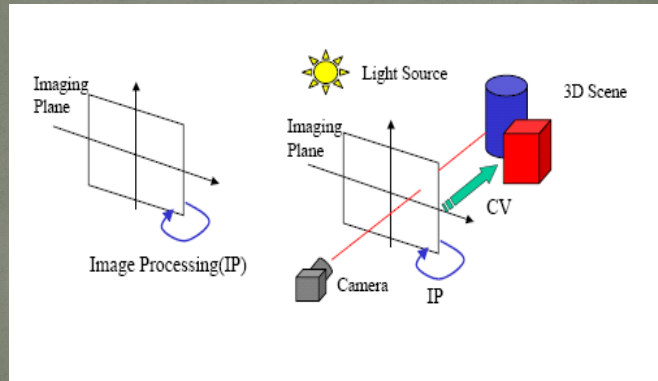
Introduction of Computer Vision

- Digital Image Processing (IP)
 - Since 1960's
 - 1970's : Recognized as a technology for practical use.
 - Most Important Issues : Image Transformation
 - Noise/Blur Compensation, Distortion Correction

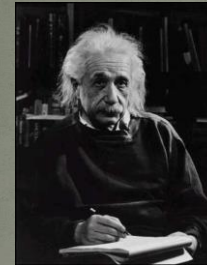
Introduction of Computer Vision

- Computer Vision (CV)
 - Emerged from AI (Artificial Intelligence) Research
 - Computational Vision proposed by Marr
 - Shape from X(X:shading, texture, contour, motion...etc.)
 - 3D Scene Understanding from 2D images
 - Robot Vision: Vision functionality for Robots
 - Media Processing, Human Interface

Image Processing / Computer Vision



Computer Vision

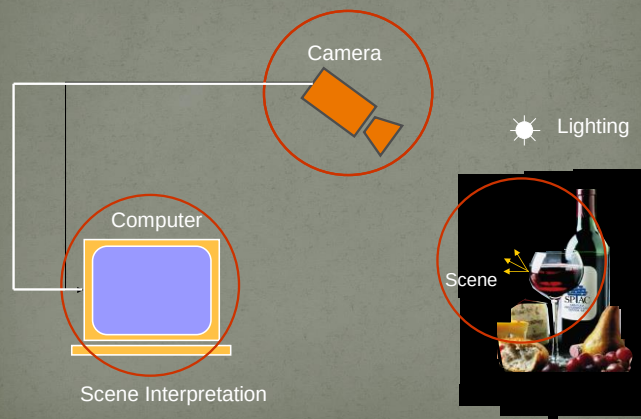


What we see

0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

What a computer sees

Components of a Computer Vision System



Example

Finding People in images

Problem 1: Given an image I

Question: Does I contain an image of a person?

“Yes” Instances



“No” Instances



Thank you

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